Franklin University - ECON 251 (Intermediate Microeconomics)
Randy Reid
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Assignment Title - "Homework Module 5"

## Problem 1

*Utility function: $u(x_1, x_2) = x_1 \cdot x_2$, with prices $p_1$, $p_2$ and income $b$.*

1. Marshallian demand functions and money illusion

Maximizing $u = x_1x_2$ subject to $p_1x_1 + p_2x_2 = b$: the tangency condition $MU_1/MU_2 = p_1/p_2$ gives $x_2/x_1 = p_1/p_2$, i.e., $p_1x_1 = p_2x_2$. Substituting into the budget constraint ($2p_1x_1 = b$) gives:

$x_1(p_1, p_2, b) = b / (2p_1)$ $\qquad$ $x_2(p_1, p_2, b) = b / (2p_2)$

The consumer is free of money illusion: scaling all prices and income by the same factor $\lambda > 0$ leaves demand unchanged, since $x_1(\lambda p_1, \lambda p_2, \lambda b) = \lambda b / (2\lambda p_1) = b / (2p_1) = x_1(p_1, p_2, b)$. Demand is homogeneous of degree zero in prices and income — only relative prices and real income matter, not the nominal price level.

2. Demand at $b = 900$, $p_1 = 25$, $p_2 = 30$

$x_1 = 900 / (2 \cdot 25) = 900 / 50 = 18$ $\qquad$ $x_2 = 900 / (2 \cdot 30) = 900 / 60 = 15$

3. Demand after $p_1$ rises to 36 ($p_2 = 30$, $b = 900$ unchanged)

$x_1 = 900 / (2 \cdot 36) = 900 / 72 = 12.5$ $\qquad$ $x_2 = 900 / (2 \cdot 30) = 15$

Good 2's quantity is unaffected by the change in $p_1$ ($x_2$ stays at 15) — a hallmark of Cobb-Douglas demand: the cross-price effect is exactly zero, so a price change in one good only affects that good's own quantity, not the other's.

4. Comment on the general Cobb-Douglas statement

For $u(x_1, x_2) = x_1^{a} \cdot x_2^{b}$ with $\alpha, \beta > 0$, the Marshallian demands are:

$x_1^* = [\alpha/(\alpha+\beta)] \cdot (b/p_1)$ $\qquad$ $x_2^* = [\beta/(\alpha+\beta)] \cdot (b/p_2)$

Expenditure on good 1 is $p_1x_1^* = [\alpha/(\alpha+\beta)] \cdot b$ — a constant fraction of income, independent of both prices, exactly as the statement claims. The precise wording needs a small correction, though: it's the expenditure share (the fraction of income spent on each good) that's fixed and price-independent, not the physical quantity purchased — quantity still falls when a good's own price rises (each good has unit own-price elasticity of demand, since $x_1^* \cdot p_1$ stays constant as $p_1$ varies). The statement is therefore correct in spirit — fixed budget shares are the defining feature of Cobb-Douglas preferences — but should be read as referring to spending shares, not quantities.

## Problem 2

1. The "money pump" argument for intransitive preferences

Suppose a consumer's preferences cycle: $A \succ B$, $B \succ C$, and $C \succ A$. Start the consumer off holding bundle A.

- Since $C \succ A$, the consumer is willing to pay a small amount $\varepsilon_1$ to trade A for C.
- Since $B \succ C$, the consumer is willing to pay a small amount $\varepsilon_2$ to trade C for B.
- Since $A \succ B$, the consumer is willing to pay a small amount $\varepsilon_3$ to trade B back for A.

After the three trades, the consumer holds exactly the same bundle (A) they started with, but has paid away $\varepsilon_1+\varepsilon_2+\varepsilon_3$ in the process. A businessman who knows the consumer's preference cycle can repeat this sequence of trades indefinitely, extracting an unbounded amount of money while the consumer's holdings never actually improve — a "money pump." This is precisely why transitivity is imposed as a rationality axiom: without it, a chain of individually "improving" trades can be fully exploited by a counterparty who sees the cycle.

2. Strict preference and indifference from the weak ordering $\succeq$

*(a) Definitions*

$$x \succ y \iff x \succeq y \text{ and not } (y \succeq x) \qquad x \sim y \iff x \succeq y \text{ and } y \succeq x$$

*(b) Do completeness and transitivity carry over?*

Completeness does not carry over: $\succeq$ is complete by assumption (any two bundles can be compared), but $\sim$ is not — two arbitrary bundles need not be indifferent to each other; they may instead stand in a strict preference relation. So indifference only relates some pairs of bundles, not all of them.
Transitivity does carry over to both $\sim$ and $\succ$, given that $\succeq$ itself is transitive. For $\sim$: if $x \sim y$ and $y \sim z$, then $x \succeq y$, $y \succeq x$, $y \succeq z$, and $z \succeq y$; transitivity of $\succeq$ applied to ($x \succeq y$, $y \succeq z$) gives $x \succeq z$, and applied to ($z \succeq y$, $y \succeq x$) gives $z \succeq x$, so $x \sim z$. For $\succ$: if $x \succ y$ and $y \succ z$, transitivity of $\succeq$ gives $x \succeq z$; if $z \succeq x$ also held, transitivity would force $z \succeq y$, contradicting $y \succ z$, so not ($z \succeq x$) — hence $x \succ z$.

3. MRS does not depend on the specific utility representation
Let $v(x_1, x_2) = f(u(x_1, x_2))$ be any strictly increasing transformation of u ($f' > 0$), representing the same preferences. By the chain rule, $\partial v/\partial x_i = f'(u)\cdot\partial u/\partial x_i$ for i = 1, 2. The marginal rate of substitution computed from v is:

$$MRS\_v = (\partial v/\partial x_1)/(\partial v/\partial x_2) = [f'(u)\cdot\partial u/\partial x_1] / [f'(u)\cdot\partial u/\partial x_2] = (\partial u/\partial x_1)/(\partial u/\partial x_2) = MRS\_u$$

The common factor $f'(u)$ cancels out of the ratio (it is strictly positive, so this is always valid), so MRS is identical under any monotonic rescaling of the utility function. MRS depends only on the shape of the indifference curves — the ordinal preference structure — not on which particular cardinal utility function is used to represent them.

4. Why marginal utility is not meaningful in ordinal utility theory
Ordinal utility theory only requires a utility function to correctly rank bundles; the numerical magnitude of utility, or of differences between utility values, carries no independent meaning. Since any strictly increasing transformation $f(u)$ represents the exact same preferences (part 3), marginal utility $MU_i = \partial u/\partial x_i$ is not invariant to such transformations — for example, replacing u with $v = 2u$ doubles every marginal utility without changing the underlying preferences, indifference curves, or choices at all. Because its numerical value depends entirely on an arbitrary choice of representation, marginal utility on its own has no well-defined economic content under ordinal utility theory. Only ratios of marginal utilities — i.e., the MRS — survive the transformation, since the arbitrary scaling factor cancels out (as shown in part 3); that is why MRS, not marginal utility itself, is the economically meaningful object in ordinal consumer theory.